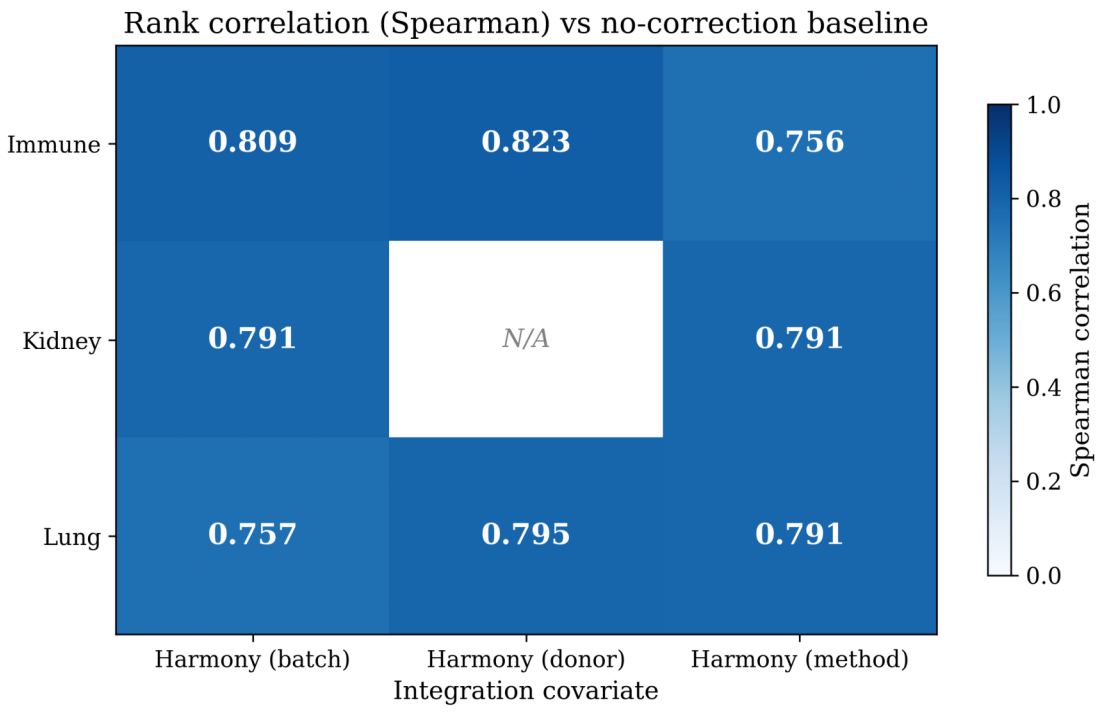
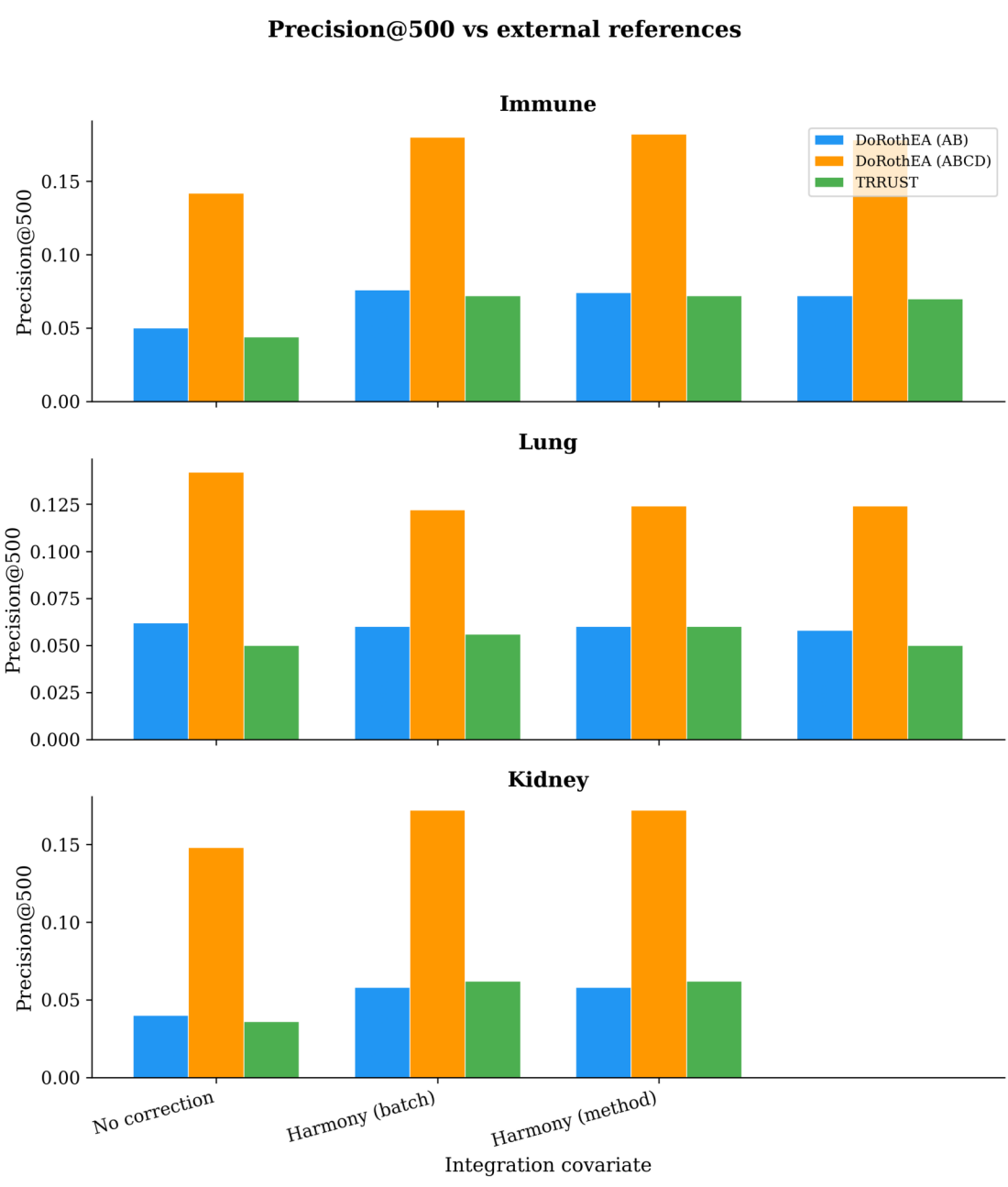
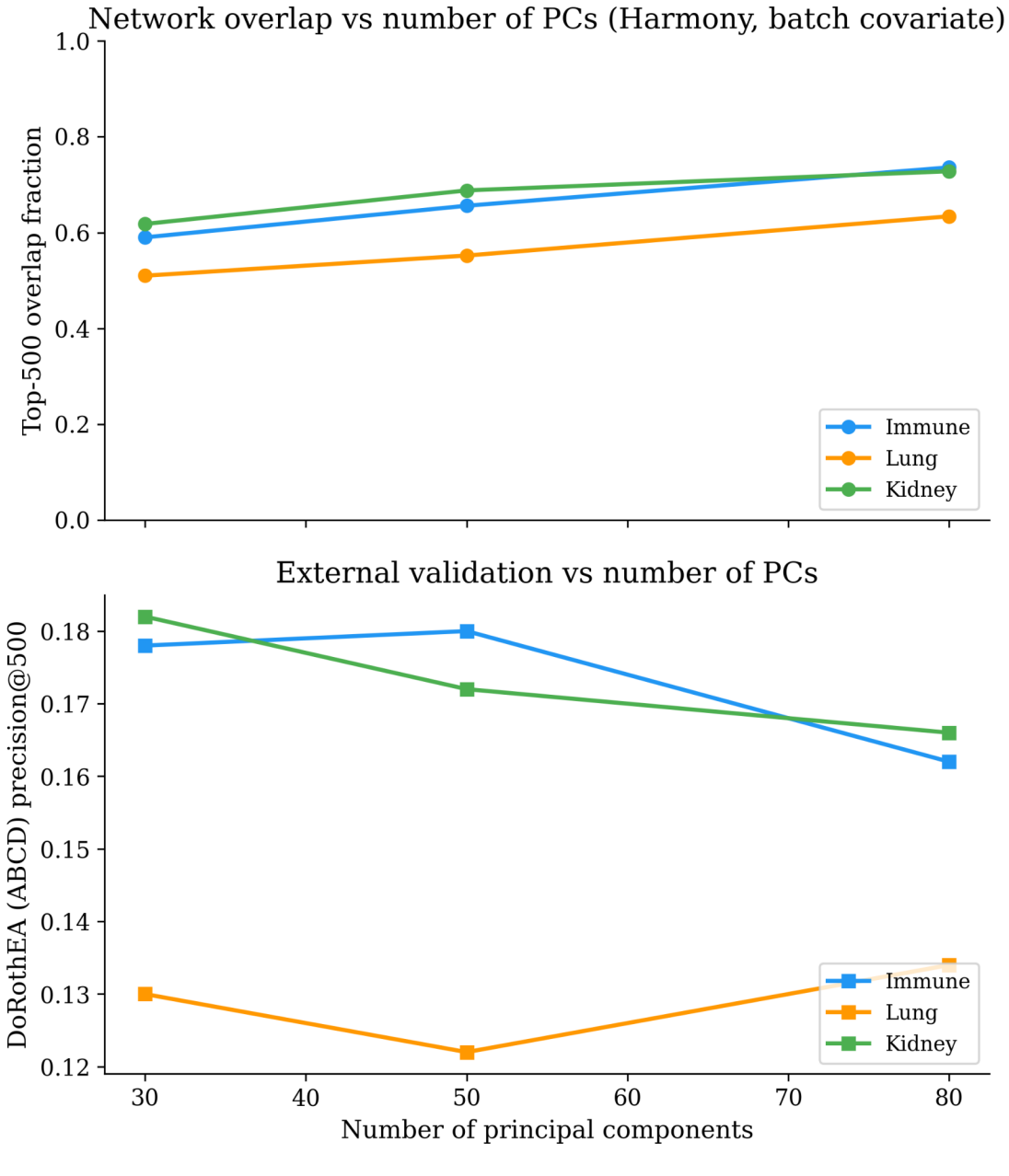
A


Rank correlation (Spearman) vs no-correction baseline
Immune
Kidney
Lung
0.809
0.823
0.756
0.791
N/A
0.791
0.757
0.795
0.791
Harmony (batch)
Harmony (donor)
Harmony (method)
Integration covariate
Spearman correlation
1.0
0.8
0.6
0.4
0.2
0.0

B


Precision@500 vs external references
Immune
DoRothEA (AB)
DoRothEA (ABCD)
TRRUST
Precision@500
0.15
0.10
0.05
0.00
Lung
0.125
0.100
0.075
0.050
0.025
0.000
Kidney
No correction
Harmony (batch)
Harmony (method)
Integration covariate

C


Network overlap vs number of PCs (Harmony, batch covariate)
Top-500 overlap fraction
1.0
0.8
0.6
0.4
0.2
0.0
Immune
Lung
Kidney
External validation vs number of PCs
DoRothEA (ABCD) precision@500
0.18
0.17
0.16
0.15
0.14
0.13
0.12
30
40
50
60
70
80
Number of principal components